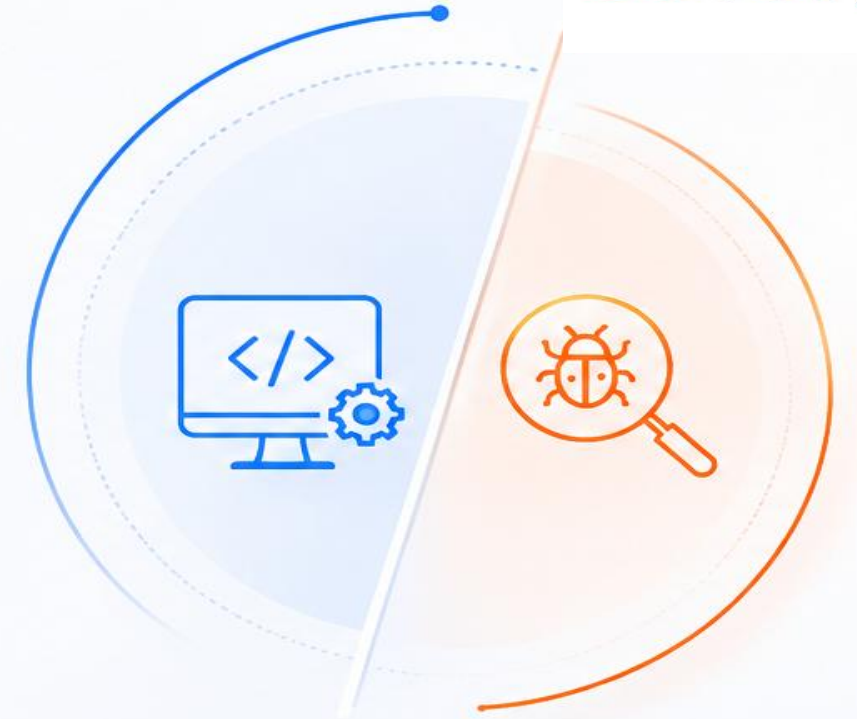


TESTING BOOTCAMP · QUALITY ASSURANCE

TESTING. QUALITY. DELIVERED.

Manual Testing (Cleartrip) & Automation Testing (SauceDemo)



Test Cases



Manual
Testing



Automation
Testing



Defect
Reporting



Selenium
Framework



PALLAB BANERJEE

Graduate Engineer Trainee

Testing Bootcamp - Batch 2026



June 2026



Internal Training Project



Automation: [SauceDemo.com](https://www.saucelabs.com)



Manual: [Cleartrip.com](https://www.cleartrip.com)

CONTENT

1:- What , Why , When ?

2:- Manual Testing

2.1:- Test Plan

2.2:- Test Scenarios

2.3:- Test Cases

2.4:- Defect repor

2.5:- Requirement Traceability Matrix

3:- Automation Testing

3.1:- Tools & Setup

3.2:- Folder Structure

3.3:- Code Files

3.4:- Allure Reports

4:- API Testing

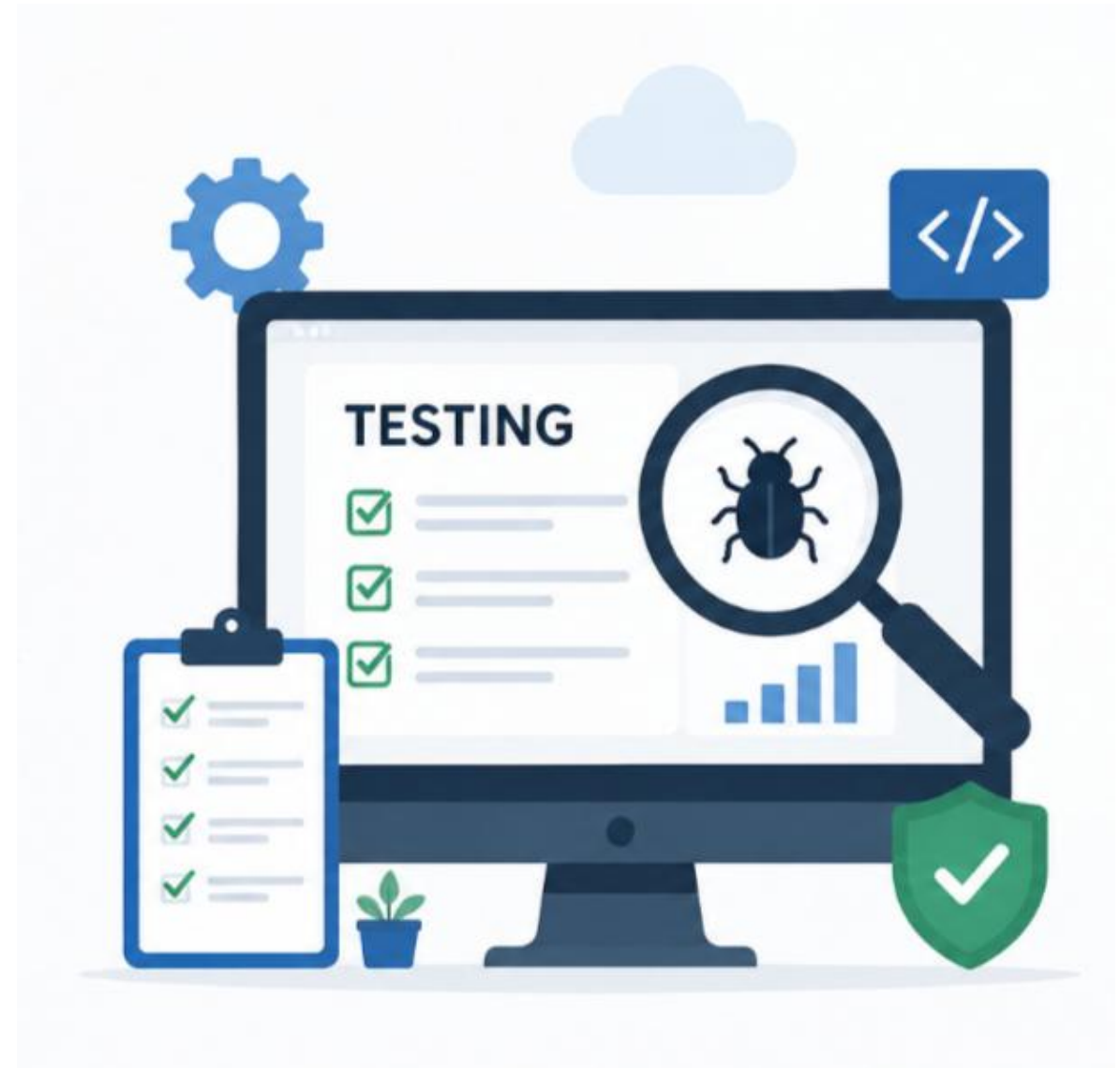
4.1 :- Postman

4.2 :- JMeter Report

5:- JDBC

5.1 :- CSV To MySQL

5.2 :- Code Files



What ?

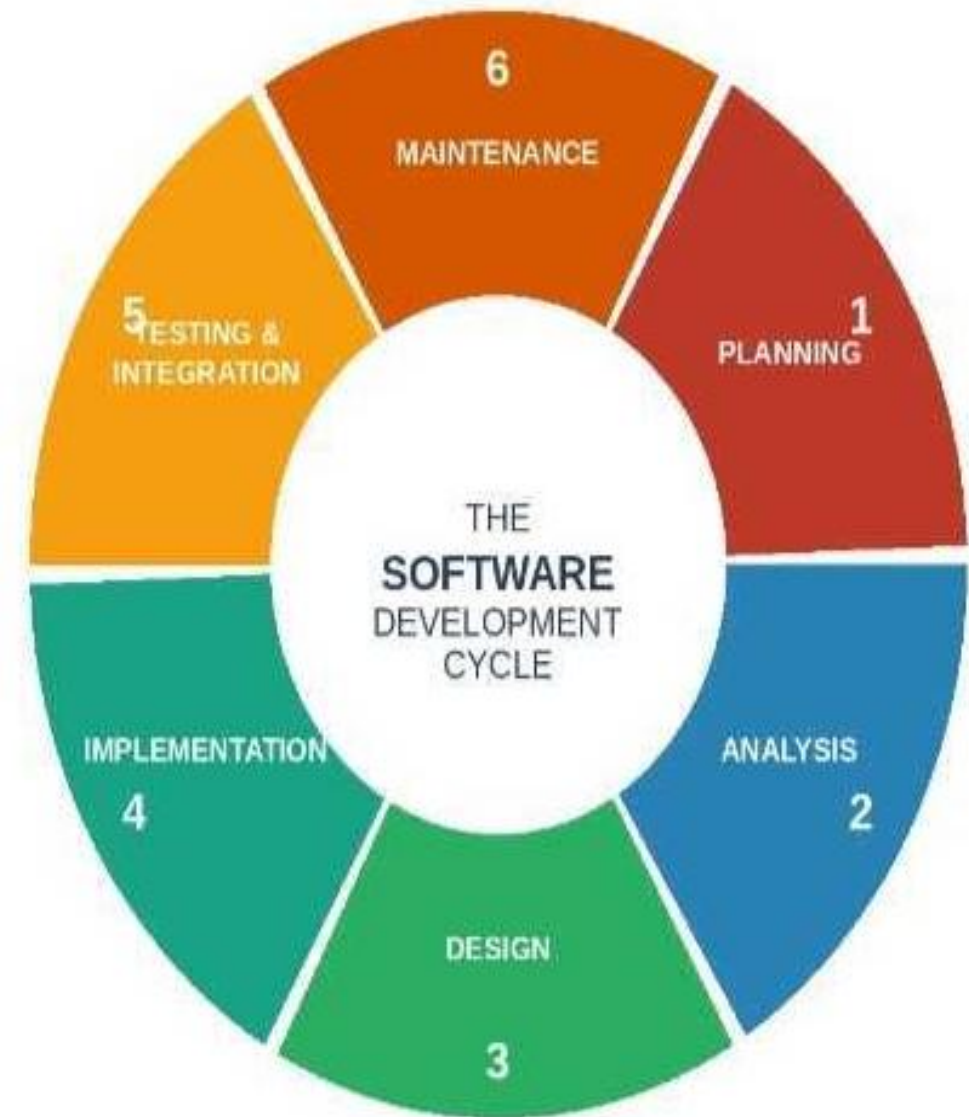
Testing is a process of execution of a program, Intention is to find a bug. The main importance of testing is to meet the quality with respect to client req.

Why ?

Testing is needed to identify defects early, ensure quality and reliability, and reduce the risk of failures that can impact users and business.

When ?

Testing is required throughout the SDLC – starting from requirement analysis and continuing through development, deployment, and maintenance.



MANUAL TESTING

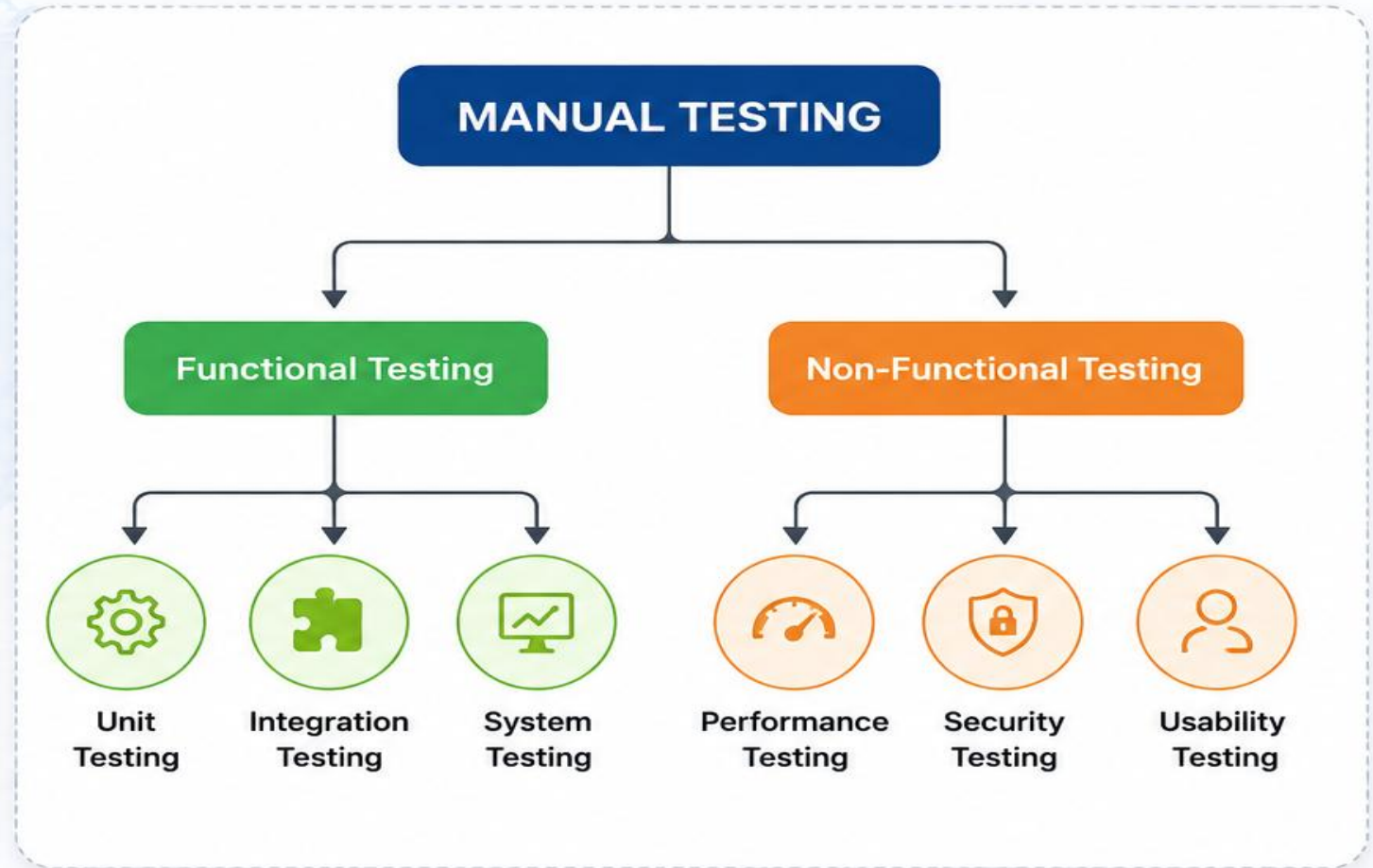
Ensuring software quality through human validation



What is Manual Testing?

Manual Testing is a software testing technique where testers execute test cases manually without using automation tools. It helps identify defects, verify functionality, and evaluate the application's usability from an end-user perspective.

- ✓ Detects UI and usability issues
- 📋 Validates business requirements
- 👤 Provides real user experience feedback
- 🔍 Useful for exploratory testing



MANUAL TESTING LIFE CYCLE (MTLC)

1. Requirement Analysis

- Understand business requirements
- Identify testable scenarios

6. Re-testing & Closure

- Verify fixes
- Perform regression testing
- Prepare test summary report

5. Defect Reporting

- Log bugs with evidence
- Assign severity and priority

4. Test Execution

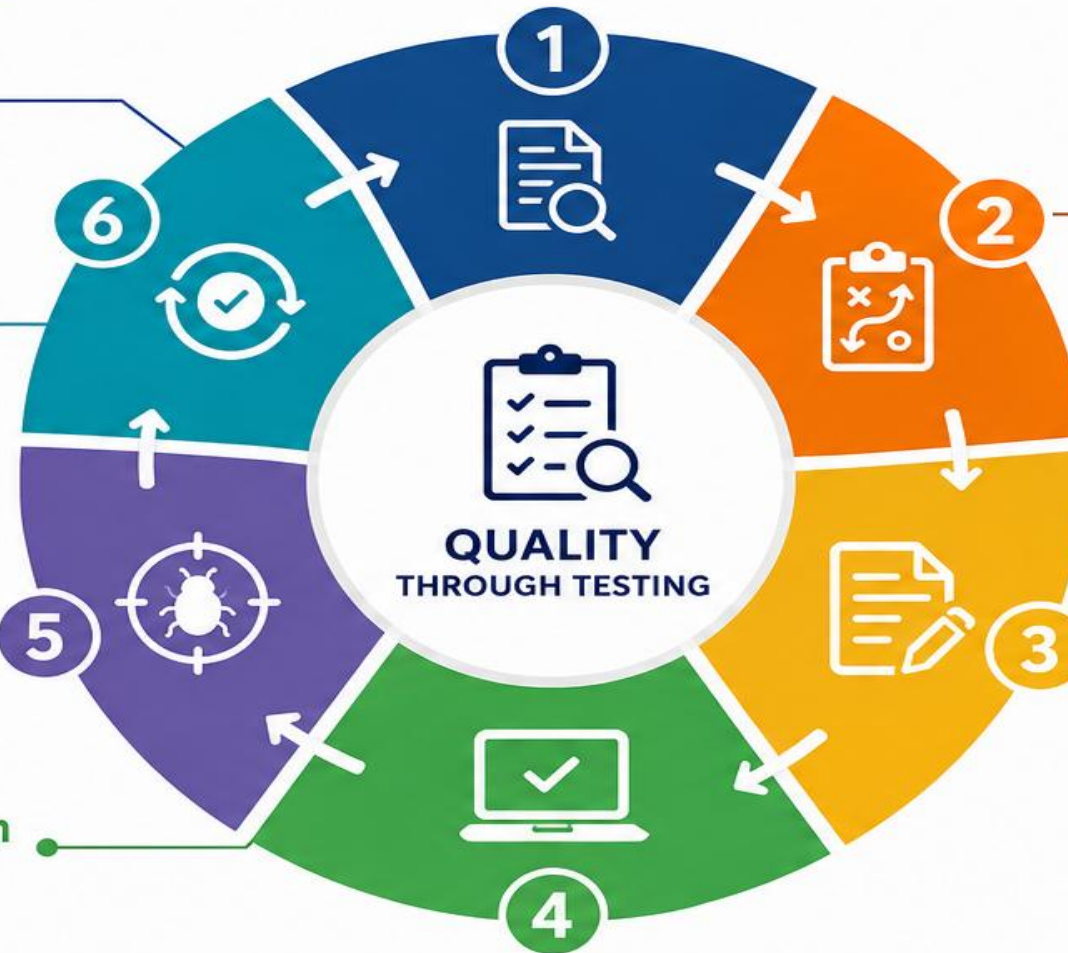
- Execute test cases
- Record actual results

2. Test Planning

- Define scope and objectives
- Estimate effort and resources

3. Test Case Design

- Create test cases and test data
- Review and update cases



KEY DELIVERABLES



Test Plan



Test Cases



Bug Reports



Test Summary Report



GOAL

Ensure the software meets requirements and is free from critical defects.

Test Plan for Cleartrip.com Website

1. Test Plan Identifier

File Name: 1.0_Cleartrip_tplan_v1.0.doc

- Document Name: Test Plan – Travel Booking Application
- Version: CT_TB_1.0
- Date: 27- May-2026
- Location: Coforge Ltd, Hyderabad

2. Document History

REV	Author	Edits	Date
1.0	Pallab Banerjee, Graduate Engineer Trainee	Initial Draft	27/05/2026

3. Introduction

Cleartrip.com is an online travel booking platform designed to help users search, compare, and book travel services such as flights, hotels, buses, and holiday packages based on preferences like destination, dates, budget, and filters. The application allows users to register, log in, manage their profiles, search for travel options, view detailed listings, and make secure bookings with flexible add-ons. The platform also provides value-added features such as ClearChoice Plus, ClearChoice Max, and “Cancel For No Reason,” enabling users to customize their booking experience with greater flexibility and convenience. This test plan defines the testing approach and scope required to validate the functional behavior, usability, performance, and reliability of the Cleartrip.com application.

4. Test Items

Specifies **what components/modules** will be tested.

Example:

- User Login Module
- OTP Verification Module
- Flight Search Module
- Flight Booking Module
- Hotel Search Module
- Hotel Booking Module
- Traveller Details Module
- Payment Gateway Module
- Booking History Module
- Logout Functionality

Medium priority features are features that have been stable and High Priority features are features that have been recently changed.

Feature to be Tested	Priority	Comments
User Login	High	
Booking History	Medium	
Flight Search	High	
Flight Booking	High	
Hotel Search	Medium	
Hotel Booking	High	
Traveller Details	High	
Payment Gateway	High	
Logout	High	

6. Features NOT to be Tested

- Help Contents and user manuals will not be tested.
- Installation and presentation will not be tested.
- Support of the other architectures also will not be tested.
- Third- party payment gateway internal logic
- Legacy modules not impacted by current release
- Experimental features

7. Test Environment

LP-HYD-85009871 HP ProBook 440 G7		Rename this PC
① Device specifications		Copy
Device name	LP-HYD-85009871	
Full device name	LP-HYD-85009871\N.COFORGETECH.COM	
Processor	Intel(R) Core(TM) i7-10510U CPU @ 1.80GHz (2.30 GHz)	
Installed RAM	16.0 GB (15.8 GB usable)	
Device ID	85BA0706-222D-4183-858D-25FEA7DEBACA	
Product ID	00329-00000-00003-AA773	
System type	64-bit operating system, x64-based processor	
Pen and touch	No pen or touch input is available for this display	



Windows specifications

Copy

Edition	Windows 11 Enterprise
Version	23H2
Installed on	22-05-2026
OS build	24H2.8871
Experience	Windows Feature Experience Pack 1000.26493.291.0
Microsoft Services Agreement	
Microsoft Software License Terms	



8. Test Approach / Test Strategy

Describes how testing will be performed.

The test strategy for ClearTrip1.0 will include manual testing with automated testing(**Selenium With Java**) where time/need permits. Automated testing will be used to test the basic Interface objects keyboard and mouse navigation or any redundant tasks best translated to automation. Automated testing will be limited to a Win-64 platform because of Automation tool limitations.

- Sanity test will include an very short battery of tests to ensure the ClearTrip1.0 runs/launches and can connect to the back end database(MySql/SQL Server 2025). Test execution will happen as individual component testing. These tests will be fully automated.
- **Acceptance tests** will include a short battery of tests that will only touch major features to ensure ClearTrip1.0 will operate smoothly at a basic level. These tests will be executed on **Windows 11 Enterprise**. The goal will be to automate all of these tests.
- Full Functional tests will include a **comprehensive test** of all features listed in this document. These tests will be limited to the **Windows 11 Enterprise**. These test will include automation with time allowing.

8.1 Types of Testing

- Smoke Testing
- Sanity Testing
- Functional Testing
- Integration Testing
- System Testing
- Regression Testing

8.2 Test Levels

- Unit Testing (by developers)
- Integration Testing
- System Testing
- Acceptance Testing

8.3 Test Techniques

- Black Box Testing
- Boundary Value Analysis
- Equivalence Partitioning
- Error Guessing **Functionality Testing**

Usability Testing

The usability testing will be accomplished by verifying the information in each window are accurate. Menus, icons and toolbar functionality will be tested as applicable to the navigation and results panes. Multi Window Overlapping will be tested because product supports opening of multiple documents.

Robustness/Reliability Testing

Test the GUI for correct keyboard and mouse navigation, windows, menus, buttons, etc. These tests will include Keyboard only navigation of ClearTrip1.0, Mouse only navigation. Correctness and existence of warning/error dialog boxes. The correct/expected functionality of buttons, icons and menus.

Stress Testing

Not applicable

9. Entry and Exit Criteria

9.1 Entry Criteria

Conditions to start testing:

- Requirements approved
- Test environment ready
- Test data available
- Build deployed

9.2 Exit Criteria

Conditions to stop testing:

- All planned test cases executed
- Critical and high defects closed
- Test summary report prepared

10. Test Deliverables

Test plan	1.0_ClearTrip_tplan.doc
Test Specification	1.0_ClearTrip_tspec.doc
Test Automation Plan	TBD()
Test Checklist	TBD
Test Logs	TBD
Test tools	Selenium 4.38, Postman, Rest Assured, Jmeter, Jenkins, Git and Git Hub and Appium and MySql Server 2025

11. Testing Tasks & Resource Needs

- Use of MySql Server database.
- Write Test Specification.
- Write Test Automation plan.
- Write Test Checklist.
- Setup workspace.
- Develop Automation script.

➤ Minimum requirements for platforms:

12. Schedule

Test Documentation

Document	Development Time		Sign-off Date	Dependencies
	Person-Days	Completion Date		
Test Plan	1 day	27/05/2026	27/05/2026	Functional Spec. Review
Test Specifications	1 day	27/05/2026	27/05/2026	Test plan, Functional Spec, Review
Test Logs	TBD	27/05/2026		
Test Summary Report	TBD	27/05/2026		
Checklist	TBD	27/05/2026		

Test Development

Deliverable	Development Time		Dependencies
	Person-Days	Completion Date	
Functional Test Suite	1	27/05/2026	
Automation build			

Test Execution

Test Activity		Execution Time (Person-Days)			Dependencies
		Alpha	Beta	FCS	
Acceptance		1 Day	1 Day	1Days	
Functional	Functionality	1	1		
	Usability	All High to Medium test cases			
	Robustness				
	Stress				
	Multi Inst.				
Regression					
Documentation Review		1	1	1	
Total Test Cycle		1	1	1	

13. Risks and Assumptions

Risks:

- **Environment Instability:** Delays in setting up the test environment could impact the schedule- VM is not working port no.-5103
- **Resource Availability:** Limited availability of testers could delay test execution.
- **Late Requirement Changes:** Changes in requirements during the testing phase could lead to additional testing efforts.
- **Contingencies:**
- **Backup Resources:** Have backup resources ready to join the testing effort if required.
- **Flexible Schedule:** Allocate buffer time in the schedule to accommodate unexpected delays.
- **Regular Communication:** Hold daily stand-up meetings to quickly address any issues that arise

14. Approvals \Sign of

POSITION	NAME	SIGNATURE	REVIEWED	APPROVAL
*Approval required Project Manager				
Project Manager			Raghavendra BN	Raghavendra BN
Client Approval	Coforge			

Test Scenarios

Scenario ID	Scenario Name	Scenario Description	Preconditions	Expected Outcome
TS_01	Verify User Login Module	Validate that the user can login to Cleartrip.com using a valid mobile number and OTP successfully.	User must have a valid registered account	User should be logged in successfully
TS_02	Flight Search Module	Validate that user can search flights using source, destination and travel date.	User is logged into the application	Relevant flight search results are displayed successfully
TS_03	Flight Booking Module	Validate that the user can book a flight successfully	User has selected an available flight	Flight booking is completed successfully
TS_04	Hotel Search Module	Validate that user can search hotels using city and dates	User is logged into the application	Relevant hotel search results are displayed successfully
TS_05	Hotel Booking Module	Validate that the user can book hotel successfully	User has selected an available hotel	Hotel booking is completed successfully
TS_06	Traveller Details Module	Validate that the user can enter traveller details correctly	User is on the Traveller Details page	Traveller details are saved successfully
TS_07	Payment Gateway Module	Validate that the user can complete payment using valid payment method	User is redirected to payment page	Payment is processed successfully
TS_08	Booking History Module	Validate that user can view previous and upcoming bookings	User is logged into the application	Booking history is displayed successfully
TS_09	Logout Functionality	Validate that user can logout from the application	User is logged into the application	User is logged out successfully

Test Cases

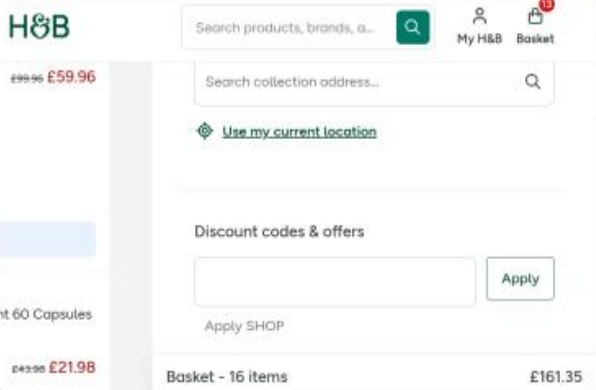
Project Name:	Travel Booking Application						
Module Name:	Cleartrip.com	Date Updated:	21-06-2026				
Last Build Tested:		Reviewed By:	Raghavendra BN				
Run No:	9 Test Cases	Executed By:	Pallab Banerjee				
Run Date:	21-06-2026						
Test Objectives:	Designing Test Cases on Cleartrip HomePage.						
Test Setup:	WPS Spreadsheet (Desktop Application)						
Req_Id	Test_Case ID	Test_Case_Description	Test_steps	Test_Data	Expected_Result	Actual_Result	Status
Req_1.0	TC_01	Validate that the user can log in to Cleartrip using a valid mobile number and OTP.	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.cleartrip.com/. 3. Verify that the Cleartrip homepage is displayed successfully. 4. User should click on the Login / Register option available on the homepage. 5. Verify that the Login page/popup is displayed successfully. 6. User should enter a valid mobile number in the required field. 7. User should click on the Get OTP button. 8. Verify that the OTP verification screen is displayed successfully. 9. User should enter the OTP received on the registered mobile number. 10. Verify that the user is logged in successfully and redirected to the Cleartrip homepage.	URL: https://www.cleartrip.com/ Mobile Number: +91 9876543211 OTP: *****	User should be logged in successfully and redirected to the homepage.	User logged in successfully and redirected to the homepage.	Passed
Req_1.1	TC_02	Validate that the user can Search flights using source, destination, and travel date.	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.cleartrip.com/ 3. Verify that the Cleartrip Home Page is displayed successfully. 4. Verify that the Flights tab is selected. 5. User should click on the "Where from?" field. 6. User should enter and select a valid source city. 7. User should click on the "Where to?" field. 8. User should enter and select a valid destination city. 9. User should click on the Departure Date field. 10. User should select a valid future travel date. 11. User should click on the "Search Flights" button. 12. Verify that the flight search results page is displayed successfully. 13. Verify that the available flights for the selected route and date are displayed	URL: https://www.cleartrip.com/ Source: Hyderabad (HYD) Destination: New Delhi (DEL) Travel Date: 25-Jun-2026	Relevant flight search results should be displayed successfully.	Relevant flight search results displayed successfully	Passed
Req_1.2	TC_03	Validate that the user can book a flight	1. User should open Microsoft Edge browser.	URL: https://www.cleartrip.com/	Flight booking should be completed	Flight booking completed	Passed

Req_1.2	TC_03	Validate that the user can book a flight successfully	1.User should open Microsoft Edge browser. 2.User should navigate to https://www.cleartrip.com/ 3.User should log in with valid credentials. 4.User should search for available flights by entering a valid source, destination, and travel date. 5.User should click on the "Search Flights" button.	URL: https://www.cleartrip.com/ Source: Kolkata (CCU) Destination: New Delhi (DEL) Travel Date: 25-Jun-2026 Traveller Name: Test User Payment Method: Credit Card	Flight booking should be completed successfully and booking confirmation should be displayed.	Flight booking completed successfully and booking confirmation displayed with booking reference number	Passed
Req_1.3	TC_04	Validate that the user can search hotels using city and check-in/check-out dates	1.User should open Microsoft Edge browser. 2.User should navigate to https://www.cleartrip.com/ 3.Verify that the Cleartrip Home Page is displayed successfully. 4.User should click on the "Hotels" tab. 5.Verify that the Hotel Search page is displayed successfully. 6.User should click on the "City, Area or Property" field. 7.User should enter and select a valid city. 8.User should click on the Check-in Date field. 9.User should select a valid check-in date. 10.User should click on the Check-out Date field. 11.User should select a valid check-out date. 12.User should select the number of rooms and guests if required. 13.User should click on the "Search Hotels" button. 14.Verify that the hotel search results page is displayed successfully. 15.Verify that available hotels for the selected city and dates are displayed.	URL: https://www.cleartrip.com/ City: New Delhi Check-in Date: 25-Jun-2026 Check-out Date: 27-Jun-2026 Guests: 2 Adults	Relevant hotel search results should be displayed successfully.	Relevant hotel search results displayed successfully	Passed
Req_1.4	TC_05	Validate that the user can book a hotel successfully	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.cleartrip.com/ 3. User should log in with valid credentials. 4. User should click on the "Hotels" tab. 5. User should search for hotels by entering a valid city, check-in date, and check-out date. 6. User should click on the "Search Hotels" button. 7. Verify that the hotel search results page is displayed successfully. 8. User should select an available hotel from the search results. 9. User should click on the "Book" or "Select Room" button. 10. Verify that the hotel details page is displayed successfully. 11. User should select a room from the available options. 12. User should click on the "Book Now" button. 13. Verify that the Traveller Details page is displayed successfully. 14. User should enter valid traveller details.	URL: https://www.cleartrip.com/ City: New Delhi Check-in Date: 25-Jun-2026 Check-out Date: 27-Jun-2026 Guest Name: Test User Payment Method: Credit Card	Hotel booking should be completed successfully and booking confirmation should be displayed.	Hotel booking completed successfully and booking confirmation displayed with booking reference number.	Passed

Req_1.5	TC_06	Validate that the user can enter traveller details correctly.	1.User should open Microsoft Edge browser. 2.User should navigate to https://www.cleartrip.com/ 3.User should log in with valid credentials. 4.User should search and select an available flight. 5.User should click on the "Book" button. 6.Verify that the Traveller Details page is displayed successfully. 7.User should enter valid traveller details such as First Name, Last Name, Gender, and Date of Birth. 8.User should enter a valid mobile number and email address. 9.User should verify all entered traveller details. 10.User should click on the "Continue" button. 11.Verify that the traveller details are accepted successfully. 12.Verify that the user is redirected to the payment page.	URL: https://www.cleartrip.com/ First Name: Test Last Name: User Gender: Male Date of Birth: 01-Jan-1999 Mobile Number: +91 9876543211 Email ID: testuser@gmail.com	Traveller details should be saved successfully and the user should be redirected to the payment page.	Traveller details saved successfully and user redirected to the payment page.	Passed
Req_1.6	TC_07	Validate that the user can view previous and upcoming bookings.	1.User should open Microsoft Edge browser. 2.User should navigate to https://www.cleartrip.com/ 3.User should log in with valid credentials. 4.Verify that the Cleartrip Home Page is displayed successfully. 5.User should click on the "My Account" option. 6.User should select the "Trips" or "My Bookings" option. 7.Verify that the Booking History page is displayed successfully. 8.User should view the list of upcoming bookings. 9.User should view the list of previous/completed bookings. 10.Verify that booking details such as Booking ID, Travel Date, Destination, and Status are displayed correctly.	URL: https://www.cleartrip.com/ Mobile Number: +91 9876543211	User should be able to view previous and upcoming bookings successfully.	Previous and upcoming bookings displayed successfully with correct booking details.	Passed
Req_1.7	TC_08	Validate that the user can complete payment using a valid payment method .	1.User should open Microsoft Edge browser. 2.User should navigate to https://www.cleartrip.com/ 3.User should log in with valid credentials. 4.User should search and select an available flight/hotel. 5.User should proceed to the Traveller Details page. 6.User should enter valid traveller details and click on Continue. 7.Verify that the Payment page is displayed successfully. 8.User should select a valid payment method (Credit Card/Debit Card/UPI/Net Banking). 9.User should enter valid payment details. 10.User should click on the "Pay Now" button. 11.User should complete the payment authentication process (OTP/UPI PIN if applicable). 12.Verify that the payment is processed successfully. 13.Verify that the booking confirmation page is displayed successfully. 14. Verify that a booking reference number is generated	URL: https://www.cleartrip.com/ Payment Method: Credit Card Card Number: XXXX XXXX XXXX XXXX Expiry Date: MM/YY CVV: XXX	Payment should be processed successfully and booking confirmation should be displayed.	Payment processed successfully and booking confirmation displayed with booking reference number.	Passed

Req_1.8	TC_09	Validate that the user can logout from the application successfully.	1.User should open Microsoft Edge browser. 2.User should navigate to https://www.cleartrip.com/ 3.User should log in with a valid mobile number and OTP. 4.Verify that the user is logged in successfully. 5.User should click on the "My Account" option available at the top-right corner of the page. 6.Verify that the account menu is displayed successfully. 7.User should click on the "Logout" option. 8.Verify that the user session is terminated successfully. 9.Verify that the user is redirected to the Cleartrip Home Page. 10.Verify that the user is no longer logged in and the Sign In/Login option is displayed.	URL: https://www.cleartrip.com/ Mobile Number: +91 9876543211	User should be logged out successfully and redirected to the Cleartrip Home Page.	User logged out successfully and redirected to the Cleartrip Home Page.	Passed
---------	-------	---	--	---	---	---	--------

Defect Report

Defect ID	Defect Description	Steps to Reproduce	Priority	Severity	Status	Created By	Assigned To	Screenshot
1	Basket count is not updated after adding or removing items from the basket .	1. Open MicrosoftEdge browser . 2. Search for https://www.hollandandbarrett.com/ 3. Open Holland &Barrett website. 4. Search for any product . 5. Click on the product and add it to the Basket. 6. Verify the Basket count is updated. 7. Navigate to the Basket page. 8. Modify the Basket by adding or removing one or more items. 9. Observe the Basket count displayed on the Basket icon. 10. Compare the displayed Basket count with the actual number of items present in the Basket.	High	Major	New	Pallab Banerjee	Raghavendra BN	
2	"Use my current location" functionality displays a blank page instead of loading nearby collection locations.	1. Open MicrosoftEdge browser . 2. Search for https://www.hollandandbarrett.com/ 3. Open Holland &Barrett website. 4. Search for any product . 5. Click on the product and add it to the Basket. 6. Go to the Click & Collect section. 7. Click Use my current location. 8. Allow location permission if prompted. 9. Observe the result. 10. A blank screen is displayed after clicking use my current location and no collection locations are shown.	High	Major	New	Pallab Banerjee	Raghavendra BN	
3	In the "Save up to Half Price" section, one product displays "Add another" while other products display "Add to basket", resulting in an inconsistent user interface.	1. Open MicrosoftEdge browser . 2. Search for https://www.hollandandbarrett.com/ 3. Open Holland &Barrett website. 4. Navigate to the Offers section. 5. Open the Save up to Half Price category. 6. Observe the button labels displayed for the products. 7. Compare the button text across the	Low	Minor	New	Pallab Banerjee	Raghavendra BN	

[Test Scenario Cleartrip](#)
[TestCase Cleartrip Homepage](#)
[Defects Holland and Barrett](#)
[RTM Case Study](#)

Requirement Traceability Matrix

Req No.	Test Case ID	Requirement Description	Status
Req_1.0	TC_01	The system should allow a registered user of Cleartrip Login Module to log in successfully using a valid mobile number and OTP.	TC_01 _ Passed
Req_1.1	TC_02	The system should allow the user of Cleartrip Flight Search Module to search flights using source, destination, and travel date.	TC_02 _ Passed
Req_1.2	TC_03	The system should allow the user of Cleartrip Flight Booking Module to book a flight successfully.	TC_03 _ Passed
Req_1.3	TC_04	The system should allow the user of Cleartrip Hotel Search Module to search hotels using city and check-in/check-out dates.	TC_04 _ Passed
Req_1.4	TC_05	The system should allow the user of Cleartrip Hotel Booking Module to book a hotel successfully.	TC_05 _ Passed
Req_1.5	TC_06	The system should allow the user of Cleartrip Traveller Details Module to enter and save traveller details correctly.	TC_06 _ Passed
Req_1.6	TC_07	The system should allow the user of Cleartrip Payment Gateway Module to complete payment using a valid payment method successfully.	TC_07 _ Passed
Req_1.7	TC_08	The system should allow the user of Cleartrip Booking History Module to view previous and upcoming bookings.	TC_08 _ Passed
Req_1.8	TC_09	The system should allow the user of Cleartrip Logout Functionality to log out successfully from the application.	TC_09 _ Passed

Peer Review



Pallab Banerjee <pallab02022003@gmail.com>

to Raghavendra ▾

Hi Raghavendra,

I have shared the test deliverables for peer review . Please find the attached the **Test Plan, Test Scenario , Test Cases , Defect Report and RTM .**

Kindly review and share your Feedback or suggestions at your convenience.

Thank you for your time and support .

Best Regards

Pallab Banerjee[85009871]

📧 12:03 (9 hours ago)



R

Raghavendra Bn

to me ▾

Hi **Pallab Banerjee**,

Thank you very much for sharing the test deliverables with me. I have successfully received all the documents, including the Test Plan, Test Scenarios, Test Cases, Defect Report, and the Requirements Traceability Matrix (RTM). I appreciate the time and effort you have invested in preparing and compiling these materials.

Over the next few days, I will be going through each of the documents in detail to ensure that everything aligns with the project requirements and quality standards. If I come across any areas that require clarification or further information, I will reach out to you directly. Additionally, if I have any suggestions for improvement or notice any discrepancies, I will make sure to communicate them promptly.

Your thoroughness in documenting the testing process is commendable, and it will certainly help streamline our review and implementation phases. Please let me know if there are any specific sections you would like me to focus on or if there are any particular concerns you have regarding the deliverables.

Once I have completed my review, I will provide you with comprehensive feedback. In the meantime, if you have any updates or additional information to share, please feel free to send them my way.

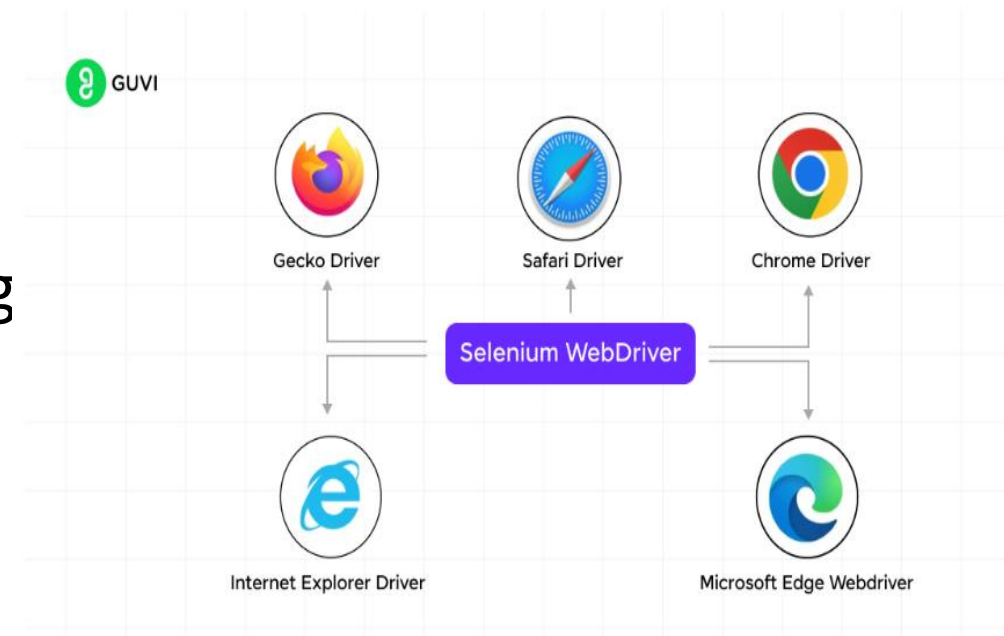
19:04 (2 hours ago)



Automation Testing

- Automation testing is a modern approach to software testing where test scenarios are executed using automated tools and scripts. Instead of manually checking each functionality, machines perform the testing process.
- Automation testing increases speed, accuracy, and reliability for repetitive and regression testing.

Selenium WebDriver



TOOLS & SETUP



JDK (Java)
24



Selenium
4.44.0



TestNG
7.11.0



Cucumber
7.30.0



Apache POI
5.4.1



Postman
Latest Version



Eclipse IDE
2025-06
Release



Allure Reports
2.35.1



Apache JMeter
5.6.3



**MySQL
Workbench**
8.0.43



Jenkins
LTS 2.516.1



JDBC



Rest Assured
5.5.5



Maven
3.9.11



Git
2.50.1

Folder Structure

- ▼ com.OpenCartDemo
 - src/main/java
 - src/main/resources
 - ▼ src/test/java
 - com.Framework.Pages
 - > BaseClass.java
 - > LoginPage.java
 - com.Framework.TestCase
 - > LoginSauceDemoTest.java
 - utility
 - > BrowserFactory.java
 - > ConfigDataProvider.java
 - > ExcelDataProvider.java
 - > Helper.java
 - src/test/resources
- > JRE System Library [JavaSE-1.8]
- > Maven Dependencies
- > allure-results
- > Configuration
 - Edge_Browser_JAR
- > ScreenShots
- > src
- > target
- > TestData
- > test-output
- pom.xml
- testng.xml



- ▼ src/test/java
 - com.Framework.Pages
 - > BaseClass.java
 - > LoginPage.java
 - com.Framework.TestCase
 - > LoginSauceDemoTest.java
 - utility
 - > BrowserFactory.java
 - > ConfigDataProvider.java
 - > ExcelDataProvider.java
 - > Helper.java



- ▼ Configuration
 - config.properties
- ▼ Edge_Browser_JAR
 - msedgedriver.exe
- > ? > ScreenShots

POM.xml

```
BaseClass.java LoginPage.java LoginSauceD... BrowseFacto... ConfigDataP... config.prop... com.OpenCar... »
1<?xml version="1.0" encoding="UTF-8"?>
2<project xmlns="https://maven.apache.org/POM/4.0.0"
3      xmlns:xsi="https://www.w3.org/2001/XMLSchema-instance"
4      xsi:schemaLocation="https://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6  <modelVersion>4.0.0</modelVersion>
7
8  <groupId>com.OpenCartDemo</groupId>
9  <artifactId>com.OpenCartDemo</artifactId>
10 <version>0.0.1-SNAPSHOT</version>
11
12 <dependencies>
13
14   <dependency>
15     <groupId>org.seleniumhq.selenium</groupId>
16     <artifactId>selenium-java</artifactId>
17     <version>4.44.0</version>
18   </dependency>
19
20   <dependency>
21     <groupId>org.testng</groupId>
22     <artifactId>testng</artifactId>
23     <version>7.12.0</version>
24     <scope>test</scope>
25   </dependency>
26
27   <dependency>
28     <groupId>au.com.bytecode</groupId>
29     <artifactId>opencsv</artifactId>
30     <version>2.4</version>
31   </dependency>
32
33   <dependency>
34     <groupId>org.apache.poi</groupId>
35     <artifactId>poi</artifactId>
36     <version>5.5.1</version>
37   </dependency>
38
39   <dependency>
40     <groupId>org.apache.poi</groupId>
41     <artifactId>poi-ooxml</artifactId>
42     <version>5.5.1</version>
43   </dependency>
44
45   <dependency>
46     <groupId>io.qameta.allure</groupId>
47     <artifactId>allure-testng</artifactId>
```

```
43
44   <dependency>
45     <groupId>io.qameta.allure</groupId>
46     <artifactId>allure-testng</artifactId>
47     <version>2.35.2</version>
48   </dependency>
49 </dependencies>
50
51 <groupId>org.apache.logging.log4j</groupId>
52 <artifactId>log4j-api</artifactId>
53 <version>2.23.1</version>
54 </dependency>
55 <dependency>
56   <groupId>org.apache.logging.log4j</groupId>
57   <artifactId>log4j-core</artifactId>
58   <version>2.23.1</version>
59 </dependency>
60 </dependencies>
61
62 </project>
```

Helper.java

```
1 package utility;
2
3 import java.io.File;
4
5 public class Helper {
6
7     public static void capturedScreenShot(WebDriver driver) {
8
9         try {
10             TakesScreenshot ts = (TakesScreenshot) driver;
11             File source = ts.getScreenshotAs(OutputType.FILE);
12
13             FileUtils.copyFile(
14                 source,
15                 new File("./ScreenShots/Screenshot_"
16                     + getCurrentDateTime() + ".png")
17             );
18
19             System.out.println("Screenshot Captured");
20
21         } catch (Exception e) {
22             System.out.println("Error while taking screenshot " + e.getMessage());
23         }
24     }
25
26     public static String getCurrentDateTime() {
27
28         DateFormat customFormat =
29             new SimpleDateFormat("MM_dd_yyyy_HH_mm_ss");
30
31         Date currentDate = new Date();
32
33         return customFormat.format(currentDate);
34     }
35 }
36 }
```

BrowserFactory.java

```
1 package utility;
2
3 import org.openqa.selenium.WebDriver;
4
5 public class BrowserFactory {
6
7     static WebDriver driver;
8
9     public static WebDriver BrowserOptions(String browserName, String url) {
10
11         if (browserName.equalsIgnoreCase("Chrome")) {
12             driver = new ChromeDriver();
13
14         } else if (browserName.equalsIgnoreCase("firefox")) {
15             driver = new FirefoxDriver();
16
17         } else if (browserName.equalsIgnoreCase("edge")) {
18             driver = new EdgeDriver();
19         }
20
21         driver.manage().window().maximize();
22         driver.get(url);
23         return driver;
24     }
25
26     public static void quitBrowser(WebDriver driver) {
27         if (driver != null) {
28             driver.quit();
29         }
30     }
31 }
32 }
```

ConfigDataProvider.java

```
1 package utility;
2
3 import java.io.File;
4
5
6 public class ConfigDataProvider {
7
8     Properties pro;
9
10     public ConfigDataProvider() {
11
12         File src = new File("./Configuration/config.properties");
13
14         try {
15             FileInputStream fis = new FileInputStream(src);
16
17             pro = new Properties();
18             pro.load(fis);
19
20         } catch (Exception e) {
21             System.out.println("Unable to load config file " + e.getMessage());
22         }
23     }
24
25     public String getBrowser() {
26         return pro.getProperty("Browser");
27     }
28
29     public String getAppURL() {
30         return pro.getProperty("AppUrl");
31     }
32 }
33 }
```

ExcelDataProvider.java

```
package utility;
import java.io.File;

public class ExcelDataProvider {

    XSSFWorkbook wb;
    DataFormatter formatter;

    public ExcelDataProvider() {

        File src = new File("./TestData/TestDataNew.xlsx");

        try {
            FileInputStream fis = new FileInputStream(src);
            wb = new XSSFWorkbook(fis);
            formatter = new DataFormatter(); // important
        } catch (Exception e) {
            System.out.println("Unable to load Excel file " + e.getMessage());
        }
    }

    public String getStringData(int sheetIndex, int row, int column) {
        return formatter.formatCellValue(
            wb.getSheetAt(sheetIndex).getRow(row).getCell(column)
        );
    }

    public String getStringData(String sheetName, int row, int column) {
        return formatter.formatCellValue(
            wb.getSheet(sheetName).getRow(row).getCell(column)
        );
    }

    public double getNumericData(String sheetName, int row, int column) {
        return wb.getSheet(sheetName)
            .getRow(row)
            .getCell(column)
            .getNumericCellValue();
    }
}
```


Allure Reports



Allure



Overview



Categories



Suites



Behaviors



Packages



Graphs



Timeline



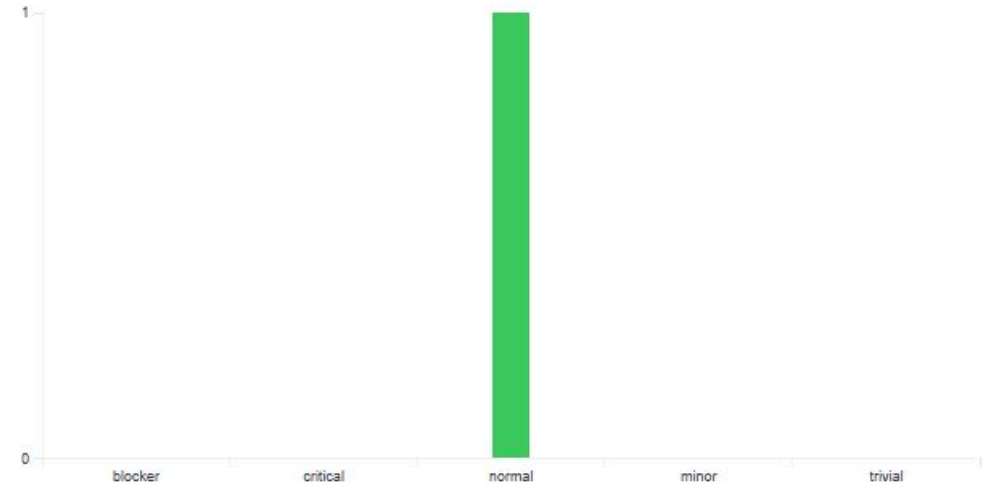
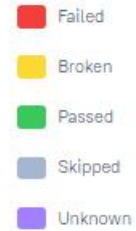
Dark theme



en Language



Collapse



Duration



Duration trend

API Testing



API Testing

- **Definition**

API Testing is the process of testing Application Programming Intertaces (APIs) to ensure they work correctly and meet expected requirements.

- Sending requests (GET, POST, PUT, DELETE, PATCH)
- Validating responses such as:
 - Status codes (200, 404, 500. etc,)
 - Response data
 - Performance

- **Why API Testing is Important**

- ✓ Checks core functiondlity of the application
- ✓ Verifies communication between systems
- ✓ Helps (dentify issues early in development
- ✓ Improves performance and speed
- ✓ Ensures data accuracy and reliablility



Postman

The screenshot displays the Postman interface with the following components:

- Top Bar:** Includes navigation icons, the workspace name "Pallab Banerjee's Workspace", a search bar, and an "Invite" button.
- Left Sidebar:** Shows a "COLLECTIONS" list with "APITestingProject" and "My Collection". Under "My Collection", there are three items: "GET Get data", "POST Post data", and "GET Get Request method". At the bottom, there are sections for "ENVIRONMENTS", "SPECS", and "FLOWS".
- APITestingProject - Run results:** A summary section showing the test was run today at 11:09:47 AM. It includes a table with the following data:

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	none	1	8s 528ms	0	0	669 ms

Below the table, there are filters for "All 0", "Passed 0", "Failed 0", "Skipped 0", "Errors 0", and a "Console log" link. There are also "List" and "Grid" view toggles.
- Test Results:** A list of five test results, each showing the method, URL, status code, response time, and size.
 - GET New Request:** https://jsonplaceholder.typicode.com/posts, Status: 200, Time: 445 ms, Size: 8.372 KB. Note: No tests found.
 - POST New Request:** https://jsonplaceholder.typicode.com/posts, Status: 201, Time: 652 ms, Size: 1.454 KB. Note: No tests found.
 - PUT New Request:** https://jsonplaceholder.typicode.com/posts/1, Status: 200, Time: 736 ms, Size: 1.348 KB. Note: No tests found.
 - PATCH New Request:** https://jsonplaceholder.typicode.com/posts/1, Status: 200, Time: 694 ms, Size: 1.462 KB. Note: No tests found.
 - DELETE New Request:** https://jsonplaceholder.typicode.com/posts/1, Status: 200, Time: 817 ms, Size: 1.249 KB. Note: No tests found.

JMeter Report

File Edit Search Run Options Tools Help



00:00:09 0 0/1

- Test Plan
 - SauceDemo - Home Page
 - Login - HTTP Request
 - Constant Timer
 - About- HTTP Request
 - Constant Timer
 - Cart - HTTP Request
 - Constant Timer
 - All Items - HTTP Request
 - Constant Timer
 - View Results Tree
 - Aggregate Graph

Aggregate Graph

Name: Aggregate Graph

Comments:

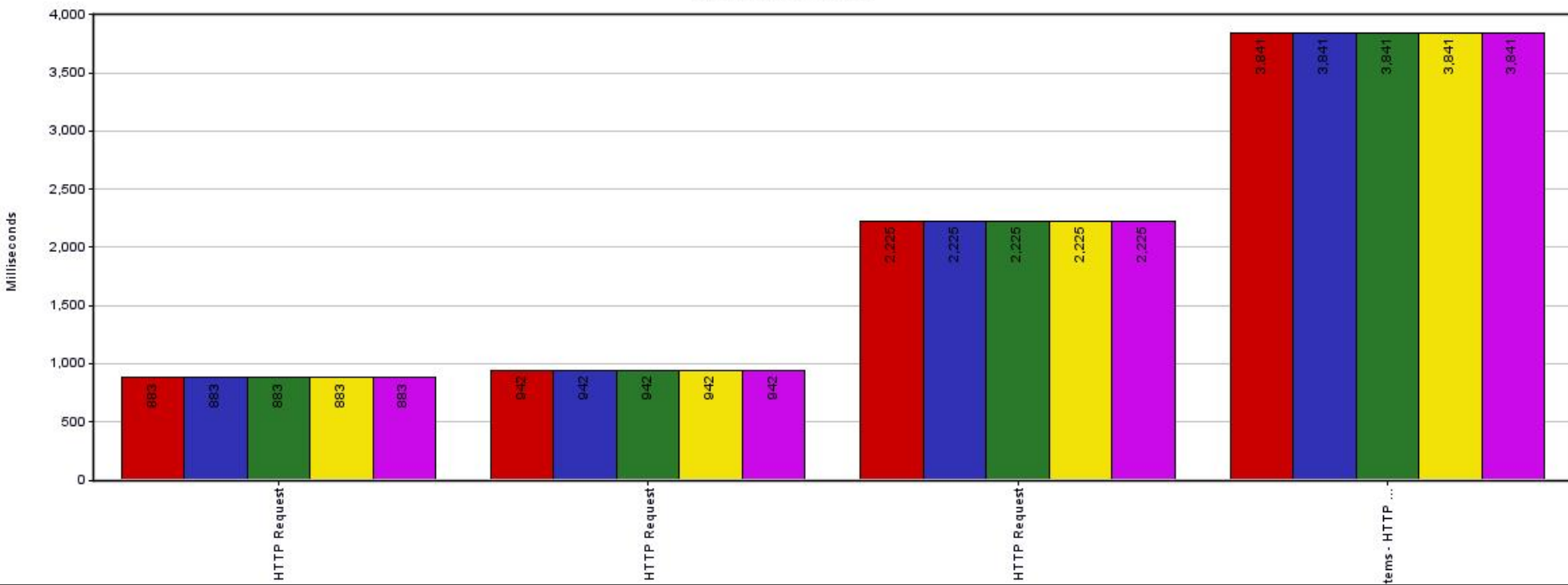
Write results to file / Read from file

Filename Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB...	Sent KB/sec
Login - HTTP Request	1	883	883	883	883	883	883	883	0.00%	1.1/sec	1.90	0.13
About- HTTP Request	1	942	942	942	942	942	942	942	0.00%	1.1/sec	59.67	0.12
Cart - HTTP Request	1	2225	2225	2225	2225	2225	2225	2225	0.00%	27.0/min	0.76	0.05
All Items - HTTP Request	1	3841	3841	3841	3841	3841	3841	3841	0.00%	15.6/min	0.44	0.03

Settings Graph

Aggregate Graph



HTML- Reports

Apache JMeter Dashboard

- Dashboard
- Charts
- Customs Graphs

Test and Report information

Source file	"SauceDemo.csv"
Start Time	"6/22/26, 12:26 PM"
End Time	"6/22/26, 12:26 PM"
Filter for display	""

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.750	500 ms	1 sec 500 ms	Total
0.500	500 ms	1 sec 500 ms	About- HTTP Request
0.500	500 ms	1 sec 500 ms	Login - HTTP Request
1.000	500 ms	1 sec 500 ms	Cart - HTTP Request
1.000	500 ms	1 sec 500 ms	All Items - HTTP Request

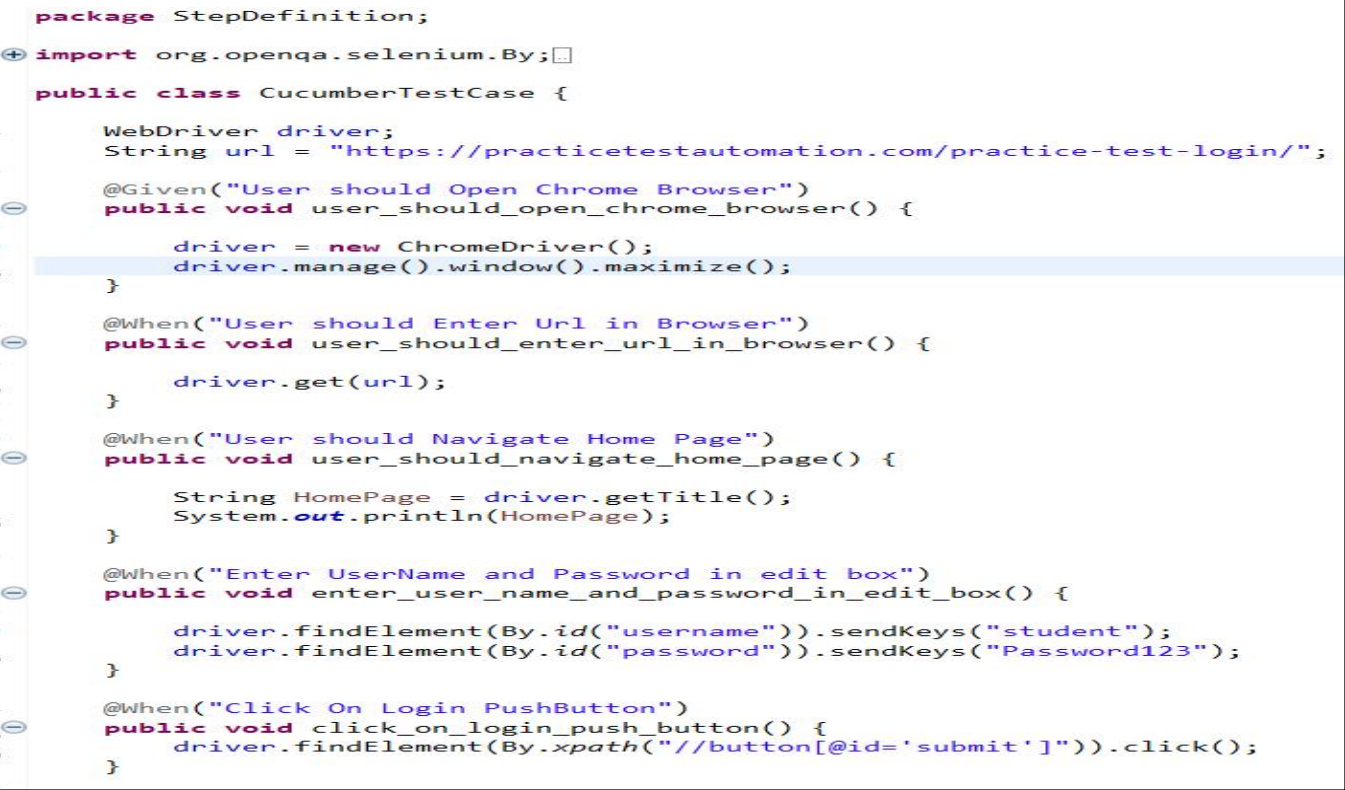
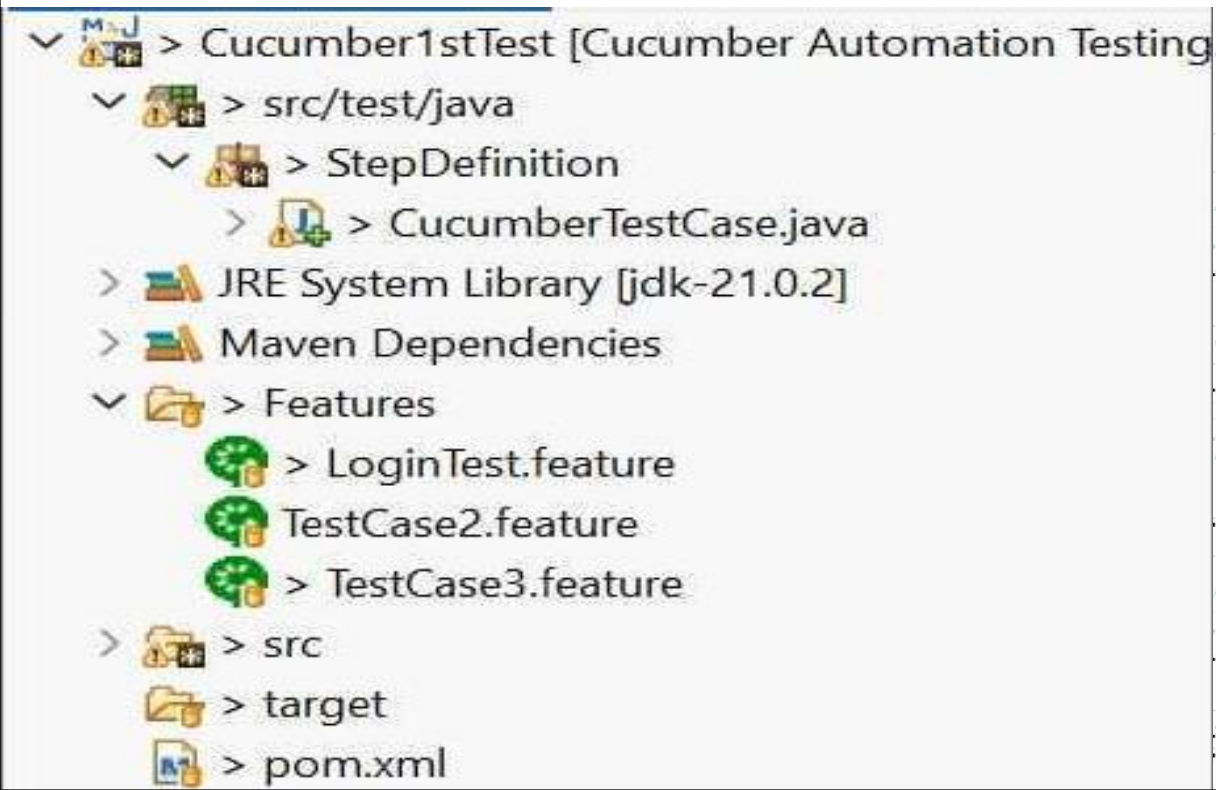
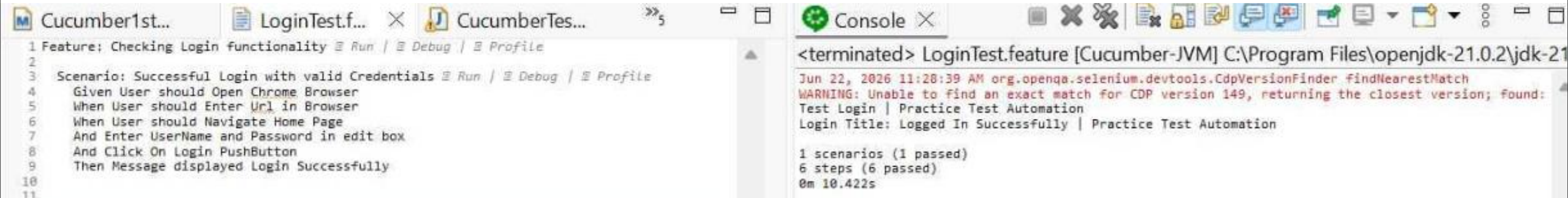
Requests Summary



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label ^	#Samples ⇅	FAIL ⇅	Error % ⇅	Average ⇅	Min ⇅	Max ⇅	Median ⇅	90th pct ⇅	95th pct ⇅	99th pct ⇅	Transactions/s ⇅	Received ⇅	Sent ⇅
Total	4	0	0.00%	547.75	80	1215	448.00	1215.00	1215.00	1215.00	1.29	19.74	0.15
About-HTTP Request	1	0	0.00%	803.00	803	803	803.00	803.00	803.00	803.00	1.25	69.99	0.14
All Items - HTTP Request	1	0	0.00%	93.00	93	93	93.00	93.00	93.00	93.00	10.75	18.05	1.25
Cart - HTTP Request	1	0	0.00%	80.00	80	80	80.00	80.00	80.00	80.00	12.50	20.98	1.45
Login - HTTP Request	1	0	0.00%	1215.00	1215	1215	1215.00	1215.00	1215.00	1215.00	0.82	1.38	0.10

Cucumber



DataDrivenTestingCSV

```
public class DataDriverTestingCSV {
    WebDriver driver;
    String CsvPath = "D:\\Workbook1.csv";

    @BeforeTest
    public void SetUp() {
        driver = new ChromeDriver();
        driver.navigate().to("https://only-testing-blog.blogspot.com/2014/05/form.html");
        driver.manage().window().maximize();
    }

    @Test
    public void CSVFile() throws Exception {
        CSVReader reader = new CSVReader(new FileReader(CsvPath));
        String[] csvCell;

        while ((csvCell = reader.readNext()) != null) {
            String FName = csvCell[0];
            String LName = csvCell[1];
            String Email = csvCell[2];
            String MNumb = csvCell[3];
            String CName = csvCell[4];

            driver.findElement(By.name("FirstName")).sendKeys(FName);
            driver.findElement(By.name("LastName")).sendKeys(LName);
            driver.findElement(By.name("EmailID")).sendKeys(Email);
            driver.findElement(By.name("MobNo")).sendKeys(MNumb);
            driver.findElement(By.name("Company")).sendKeys(CName);

            Thread.sleep(8000);
            driver.findElement(By.xpath("//input[@value='Submit']")).click();
            Thread.sleep(8000);
            driver.switchTo().alert().accept();
        }
    }

    @AfterTest
    public void endTest() {
        driver.quit();
    }
}
```


RestAssured

The screenshot displays the Eclipse IDE interface with the following components:

- Package Explorer:** Shows the project structure for 'rest.assured', including 'src/main/java', 'src/main/resources', 'src/test/java', 'src/test/resources', 'JRE System Library [JavaSE-1.8]', 'Maven Dependencies', 'src', 'target', 'test-output', 'pom.xml', and 'testng.xml'.
- Editor:** Displays the 'CompletePutPostTest.java' file with the following code:

```
1 package com.example;
2
3 import io.restassured.RestAssured;
4
11
12 Run ALL
13 public class CompletePutPostTest {
14     private static final String BASE_URL = "https://jsonplaceholder
15     private JSONObject testUser;
16     private int createdUserId;
17
18     @BeforeClass
19     public void setup() {
20         RestAssured.baseURI = BASE_URL;
21         // Common test data
22         testUser = new JSONObject();
23         testUser.put("name", "Test User");
24         testUser.put("email", "raghu.astepahead@gmail.com");
25         testUser.put("username", "raghavendra");
26     }
27 }
```
- Problems:** No issues are listed.
- Results of running class CompletePutPostTest:** Shows the test execution summary:
 - Search:
 - Passed: 5
 - Failed: 0
 - Skipped: 0
 - ests: 1/1 Methods: 5 (6393 ms)
- Test Results:** A tree view showing the test suite and individual test methods:
 - Default suite (5/0/0/0) (5.638 s)
 - Default test (5.638 s)
 - com.example.CompletePutPostTest
 - testPostRequestComplete (2.278 s)
 - testPutRequestComplete (0.724 s)
 - testPostRequestBDDComplete (1.068 s)
 - testPutRequestBDDComplete (0.9 s)
 - testPutAndPostWithExtraction (0.668 s)
- Failure Exception:** No exceptions are listed.
- Console:** Displays the test output:

```
<terminated> CompletePutPostTest [TestNG] C:\Users\Bappi.Gorain\.p2\pool\plugins\org.e
===== POST REQUEST - TRADITIONAL =====
Request Body: {
  "name": "Test User",
  "id": 1,
  "email": "raghu.astepahead@gmail.com",
  "username": "raghavendra"
}

--- POST Response Details ---
1. Status Code: 201
2. Complete Response: {
  "name": "Test User",
  "id": 11,
  "email": "raghu.astepahead@gmail.com",
  "username": "raghavendra"
}
```

[illegible]

22	06-06-2026	TestNGDifferentBrowser	https://github.com/codewithpallab/TestNgDifferentBrowser							
23	06-06-2026	SynchronizationSelenium	https://github.com/codewithpallab/SynchronizationSelenium							
24	08-06-2026	JMeter-CaseStudy1	https://github.com/codewithpallab/J-Meter-Case-Study1							
25	08-06-2026	JMeter-CaseStudy2	https://github.com/codewithpallab/J-Meter-Case-Study2							
26	09-06-2026	JMeterPracticeExample	https://github.com/codewithpallab/J-Meter-Example							
27	09-06-2026	ChasePage-CaseStudy	https://github.com/codewithpallab/ChasePage							
28	10-06-2026	ProfileBrightest-CaseStudy	https://github.com/codewithpallab/ProfileBrightest							
29	10-06-2026	PracticeAutomationTest-CaseStudy	https://github.com/codewithpallab/PractiseTestAutomation							
30	12-06-2026	RestAssure--	codewithpallab/RestAssuredTest-19June							
31	14-06-2026	Selenium-CaseStudy-1	https://github.com/codewithpallab/SeleniumTestCases							
32	14-06-2026	Selenium-Case-Study-2	https://github.com/codewithpallab/SeleniumTestCases							
33	14-06-2026	Selenium-Case-Study-6	https://github.com/codewithpallab/SeleniumTestCases							
34	14-06-2026	Selenium-Case-Study-7	https://github.com/codewithpallab/SeleniumTestCases							
35	14-06-2026	Selenium-Case-Study-8	https://github.com/codewithpallab/SeleniumTestCases							
36	14-06-2026	Selenium-Case-Study-9	https://github.com/codewithpallab/SeleniumTestCases							
37	14-06-2026	Selenium-Case-Study-10	https://github.com/codewithpallab/SeleniumTestCases							
38	14-06-2026	Selenium-Case-Study-11	https://github.com/codewithpallab/SeleniumTestCases							
39	14-06-2026	Selenium-Case-Study-12	https://github.com/codewithpallab/SeleniumTestCases							
40	14-06-2026	Selenium-Case-Study-13	https://github.com/codewithpallab/SeleniumTestCases							
41	14-06-2026	Selenium-Case-Study-14	https://github.com/codewithpallab/SeleniumTestCases							
42	14-06-2026	Selenium-Java-Automation	https://github.com/codewithpallab/PostManAssignment							
43	17-06-2026	Cucumber Case Study	https://github.com/codewithpallab/SeleniumCucumber							
44	19-06-2026	POSTMAN---CaseStudy	https://github.com/codewithpallab/PostManAssignment							

Thank you !